

## **1. What is a Computer Network? Why do we need it?**

### **Follow-up Questions:**

- Give some real-life examples of computer networks.
  - What are the advantages of networking?
  - What is the Internet?
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## **2. What is the difference between LAN, MAN, and WAN?**

### **Follow-up Questions:**

- Which one covers the largest geographical area?
  - Give one practical example of each.
  - Which network do colleges generally use?
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## **3. Explain the OSI Model.**

### **Follow-up Questions:**

- Name all seven layers.
  - Which layer is responsible for routing?
  - Which layer is responsible for encryption?
  - Which layer does HTTP work on?
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## **4. What is the TCP/IP Model? How is it different from the OSI Model?**

### **Follow-up Questions:**

- How many layers are there in TCP/IP?
  - Why is TCP/IP more commonly used?
  - Which protocols belong to each layer?
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## **5. What is the difference between TCP and UDP?**

### **Follow-up Questions:**

- Which protocol is connection-oriented?

- Which protocol is faster?
  - Give two real-world applications of TCP and UDP.
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## **6. What is an IP Address?**

### **Follow-up Questions:**

- Difference between IPv4 and IPv6.
  - What is a Private IP Address?
  - What is a Public IP Address?
  - What is a Loopback Address?
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## **7. What is the difference between a Hub, Switch, and Router?**

### **Follow-up Questions:**

- Which device works at Layer 2?
  - Which device forwards packets using IP addresses?
  - Which device creates collision domains?
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## **8. What is DNS? What is DHCP?**

### **Follow-up Questions:**

- Why is DNS required?
  - What happens when you type a website URL into the browser?
  - What does DHCP automatically assign?
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## **9. What is HTTP and HTTPS?**

### **Follow-up Questions:**

- Which port does HTTP use?
- Which port does HTTPS use?
- Why is HTTPS more secure?
- What is SSL/TLS?

### 10. Scenario-Based Question

A user reports that they are connected to Wi-Fi, but websites are not opening.

**Questions:**

- What could be the possible reasons?
- How would you troubleshoot the issue step by step?
- Which commands would you use to diagnose the problem?
- How would you check whether the issue is related to DNS, IP configuration, or Internet connectivity?